

Artificial Intelligence

Assignment: Designing and Evaluating an AI Agent

Personalized Book Recommendation Agent
(Rule-Based AI Agent)

Submitted By: Huzaifa Haroon

Seat No: 23122066

Section: Section B

Personalized Book Recommendation Agent

(Rule-Based AI Agent)

1. Problem Identification in Terms of an Agent

Readers today have access to millions of books, but not every book in a favorite genre suits their current mood, age group, or reading history, and manually checking this for every option takes time; this problem fits an intelligent agent because it needs a system that can sense the reader's details and the candidate book (its environment), reason over this information using a set of predefined preference rules, and act by producing a recommendation that helps the reader decide whether to pick up the book. The agent designed here, called the **Personalized Book Recommendation Agent**, is a rule-based agent: it does not learn from data, but works with a fixed knowledge base of IF-THEN rules built around genre preference, mood, age suitability, and previous reading history, checking a reader's profile and a candidate book against these rules to produce a simple, personalized recommendation with reasoning.

2. Problem Formulation

Input: Reader's age, favorite genre, current mood, recently read book, and the candidate book.

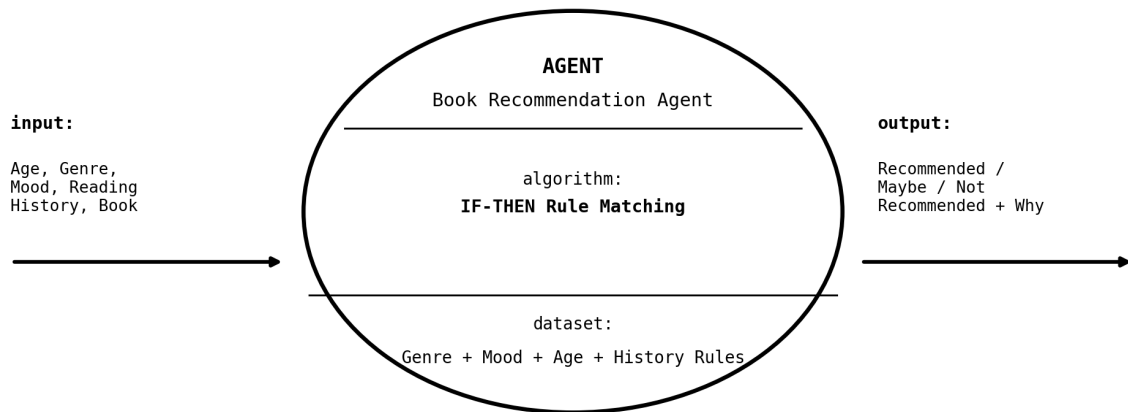
Dataset: A set of predefined recommendation rules built from genre match, mood suitability, age group, and reading history.

i.e.,

Age	Favorite Genre	Mood	Recently Read	Candidate Book	Rule / Condition	Recommendation
20	Fantasy	Adventurous	The Hobbit	Mistborn	Genre and mood both match	Highly Recommended
30	Romance	Relaxed	Pride and Prejudice	Me Before You	Genre matches, mood fits	Recommended
45	Horror	Adventurous	None	The Shining	Mood does not match genre	Maybe
16	Fantasy	Sad	Harry Potter	A Court of Thorns and Roses	Genre and age group match	Recommended
60	Non-fiction	Curious	Sapiens	Educated	Strong genre and topic match	Highly Recommended

Algorithm: Rule-Based AI using IF-THEN matching, where the reader's input is compared against the rules in the dataset to reach a decision.

Output: Highly Recommended / Recommended / Maybe / Not Recommended, along with a short reason.



3. Evaluation in the Context of PEAS

PEAS stands for Performance measure, Environment, Actuators, and Sensors. It is a standard way of describing the task environment of an agent.

PEAS Element	Description for the Book Recommendation Agent
Performance Measure	How relevant the recommendation is to the reader, overall reader satisfaction, correctness of matching, and how quickly the suggestion is produced.
Environment	The reader's personal profile (age, favorite genre, mood, reading history), the book database, and the interface through which the reader enters information and views results.
Actuators	The output/display module that shows the recommendation label (Highly Recommended / Recommended / Maybe / Not Recommended) along with a short reason.
Sensors	The input module that reads the reader's entered or stored details: age, favorite genre, current mood, recently read books, and the book being queried.

4. Environment Properties

The task environment of the Personalized Book Recommendation Agent can also be looked at through the standard dimensions used to describe intelligent agents. Each of these is explained below.

Observability: Partially observable

The agent can only work with the details a reader chooses to give it, such as favorite genre, mood, and recently read books. It has no way of sensing things the reader has not entered, like their true taste at that exact moment or books they have read elsewhere without recording them,

so its view of the reader is incomplete.

Determinism: Deterministic within the rule base

Since the agent is rule-based, giving it the same reader profile and the same candidate book will always trigger the same rules and produce the same recommendation. There is no randomness involved in how it reaches its decision.

Episodic vs. Sequential: Episodic

Each book query is handled as its own separate episode. The recommendation given for one book does not depend on what was recommended earlier for a different book, even though the reader's saved profile stays the same across queries.

Static vs. Dynamic: Semi-static (mostly static)

While the agent is working out a single recommendation, the reader's profile and the book database stay fixed. Changes only happen between sessions, for example when the reader updates their mood or finishes a new book, so the environment is treated as static during each individual query.

Discrete vs. Continuous: Discrete

Genres, moods, age groups, and recommendation levels are all represented as a limited set of fixed categories rather than continuous values, and the rules are written to match these discrete categories.

Known vs. Unknown: Known

The rules connecting genre, mood, age, and reading history to a recommendation are decided in advance by the designer based on general reading patterns, so the agent already knows how its environment behaves before it starts working.

Single-agent vs. Multi-agent: Single-agent

The system works as a single personal recommendation assistant for one reader at a time. It does not need to cooperate or compete with any other independent agent while making its decision.